

CHAPTER 1

Tang
&
Rong

Rong
Tang

2019–
2025

An-
nals
of
Sta-
tis-
tics

MCMC

1.

1.1.

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 D GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

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Tang
&
Yang
2019–
2025

MCMC

1

$\mathcal{Z}$
VAE

$\mathcal{Z}$

MCMC

2.

2.1.

.

$$\mathcal{M}^*$$

$$\mathbb{R}^D$$

$$d$$

$$\beta\text{-}$$

$$\beta \geq 1$$

$$P^*$$

$$\mathcal{M}^*$$

$$\mathrm{vol}_{\mathcal{M}^*}$$

$$p^*$$

$$\alpha-$$

$$\alpha \in [0,\beta-1]$$

$$p^*$$

$$\mathbb{R}^{\mathcal{D}}$$

Lebesgue

Lebesgue

Par-
ti-
tion
of
Unity
Tang

$$(U_\lambda, \varphi_\lambda)$$

$$\{\gamma_\lambda\}$$

Tang
(2022)

Tang

(2023a)

Reach

reach

$$\text{reach}(\mathcal{M}) = \inf\{r > 0 : \mathcal{M}$$

r -

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu, \nu) = \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\mu - \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\nu \right|.$$

$\mathcal{F}$ γ -
Hölder

Sobolev

1-
LipschitzWasserstein-
1

RKHS

MMD
 γ

trade-
off

$$\gamma = 1$$

$$d_\gamma$$

Wasserstein-
1

$$\gamma \rightarrow 0^+$$

$$d_\gamma$$

Tang
(2025a)

2.3.

•
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$$n^{-1/2}$$

$$d$$

β -

d

Haus-
dorff

D

Lep-
ski

chain-
ing
peel-
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lo-
cal-
iza-
tion

Gir-
sanov

Poincaré

2.4.

·
Gauss-
ian

Langevin

$$p_{\sigma}(x) = \int p_{\text{data}}(y) \phi_{\sigma}(x - y) \, \mathrm{d}y$$

$$L_{\infty}$$

-

$$\nabla \log p_t(x)$$

$$t$$

$$t =$$

$$0$$

$$\frac{t}{T} =$$

Tang
(2024a)

$$\mathcal{S}_{\text{NN}}$$

$$t$$

Girsanov

Gir-
sanov

SDE

3.

3.1. Tang
&
Yang
(2022)
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Rong
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“Min-
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2022.

 D GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

α

β

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par-
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of
unity

VAE

Tay-
lor

Lep-
ski

Fano

Varshamov-
Gilbert

Tang
(2022)

Tang
(2023a,
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3.2. Tang
&
Yang
(2023a)

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Rong
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ICML,
2023.

VAE
GAN

MLD-
MAE

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ρ_k

1-
Wasserstein

$$\alpha-$$

$$\beta-$$

$$d$$

$$O\Big(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\Big)\vee O\Big(\frac{\log n}{\sqrt{n}}\Big),$$

$$\begin{array}{l} \alpha\leq \\ \beta- \\ 1 \end{array}$$

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Tang
(2022)

Tang
(2022)

3.3. Tang
et
al.
(2023b)

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Rong
Tang,
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JRSS
B,
2023.

Stiefel

Un-
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tainty
Quan-
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ca-
tion

Bayesian

**RPE-
TEL**

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(2)

(3)

Bernstein-
von
Mises
(BvM)

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(RRWM)

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**3.4. Tang
&
Yang
(2023c)**

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Rong
Tang
and
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“Min-
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Two-
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Test
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Losses,”
*AIS-
TATS*,
2023.

MMD

$O(n^2)$

Besov

Besov

1

ℓ_1
 ℓ_2

Wasser-
stein

1

2

 $O(n \log n +$

$$n^{2d/(4\alpha+d)})$$

MMD

$$O(n^2)$$

**3.5. Tang
&
Yang
(2024a)**

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Rong
Tang
and
Yun
Yang.
“Adap-
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Struc-
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AIS-
TATS,
2024.

SOTA

Langevin

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d D
LangevinGauss-
ian $\tilde{O}(\sigma)$

σ n d

L_∞

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1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

Tang
(2022)

Tang
(2022)

3.6. Tang (2024b)

Metropolis-
Adjusted
Langevin

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Rong
Tang
and
Yun
Yang.
“Metropolis-
Adjusted
Langevin
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Di-
men-
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De-
pen-
dency
and
Be-
yond.”
ICML,
2024.

MALA

MCMC

$$O(d^{1/3})$$

s-
conductance
pro-
file

MALA

$$O(d^{1/3})$$

burn-
in

s-
conductance
pro-
file

con-
duc-
tance

Chen
et
al.
(2020)

con-
duc-
tance
pro-
file

warm-
start

$\log M_0$

$\log \log M_0$

Tang
(2023b)

Tang
(2023b)

BvM

BvM

MALA

3.7. Tang
&
Yang
(2025a)

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Rong
Tang
and
Yun

Yang.
“Min-
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Man-
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arXiv,
2025.

X

Y
 $\mu_{Y|X}^*$
 X
 Y

IPM

(1)
Regime
1

Lebesgue

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

(2)
Regime
2

:

$$\mathcal{M}_{Y|X}$$

$$X$$

$$\begin{array}{l} n^{-\gamma\beta_Y/d_Y} \\ (3) \end{array}$$

Regime
3

:

$$\mathcal{M}_{Y|x}$$

$$x$$

$$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$$

$$\begin{array}{l} \gamma\text{-} \\ \text{H\"older} \\ \text{IPM} \\ d_\gamma \end{array}$$

$$\begin{array}{l} \gamma = \\ 0 \end{array}$$

$$\begin{array}{l} 1\text{-} \\ \text{Wasserstein} \end{array}$$

$$\gamma = 1$$

**3.8. Tang
&
Yang
(2025b)**
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Rong
Tang
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Yun
Yang.
“Con-
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Minimax-
Optimal
and
Manifold-
Adaptive,”
arXiv,
2025.

Tang
(2024a)

(1)

$$\alpha_X \in [0, 1]$$

(2)

X

Y

Wasser-
stein

(d_X, d_Y)

(D_X, D_Y)

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Gir-
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Tang

**(2024a,
2025a)**

Tang
(2024a)

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Tang
(2025a)

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4.

Tang-
Yang

2019–
2025

4.1.

•
Tang (2022) $\longrightarrow$ Tang (2023a) MLDMAE $\longrightarrow$ Tang (2025a)

4.2.

VAE

•
 VAE (Tang 2021) $\longrightarrow$ Tang (2024a) $\longrightarrow$ Tang (2025b)

Tang
(2024a)

$t >$
0

Lebesgue

4.3.

MCMC.

Tang (2023b) BvM + RRWM $\xrightarrow{\text{BvM}}$ Tang (2024b) MALA

BvM

MALA

$$d^{1/3}$$

5.

5.1.

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(1)

2021

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Tang
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Yang,
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Tang
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(3) **2023a**

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Tang
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(4) **2023b**

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Tang
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(5) **2023c**

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Tang
&
Yang,
AIS-
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MALA

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5.2.

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**Par-
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Unity**
Tang
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2023a)

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γ -
Hölder
IPM
Tang
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2023c,
2025a)

(3) **Gaussian**

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Tang
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Langevin

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**Bernstein-
von
Mises**

Tang
et
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(2023b)

BvM

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(7) **Girsanov**

Tang
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Gir-
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Tang-
Yang

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